



DEPARTMENT OF COMPUTER SCIENCE

TDT4215 - RECOMMENDER SYSTEMS

Project Report - Group 11

Authors:

Larsen, Frikk Ormestad
Mahmoud, Mehdi Mohamed
Saeidmehr, Maryam
Sørestad, Svein Kåre

Date: 22.04.2026

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1 Introduction

Recommender systems have become a central component of many digital platforms, and their growing impact has driven sustained research interest. Several different families of recommender systems have emerged as the standard, most notably strategies like content-based filtering and collaborative filtering. Understanding the strengths and weaknesses of these systems is crucial for selecting the right approach for a given application. This report documents the development and comparison of several recommender systems for news, as part of a group project. The goal of the project was to gain experience with recommender systems, and to investigate how different recommendation strategies compare to each other. The MIND dataset was chosen due to its widespread use in the field to benchmark recommendation strategies [1]. In the following sections, the theoretical background of the implemented strategies is briefly presented, followed by the exploratory data analysis, methodology and results.

2 Theory

This section presents the necessary theoretical background to understand the implementations and results in the study.

2.1 Content-Based Filtering

Content-based filtering is a recommendation strategy where items similar to the ones liked by a user are recommended. Items are often represented as vectors, where each feature is some attribute of the item. These features are often things like categories, genre and keywords. When the user interacts with items, the features of the items are used to generate a user profile, often by averaging or weighting the item vectors. A similarity measure is then defined, and used to calculate a similarity between the user profile and potential recommendations. One large benefit of a content-based filtering approach is that the system can easily handle the cold-start problem, due to it being independent from ratings by the user [2].

2.2 Collaborative Filtering

Collaborative filtering (CF) is a recommendation approach that predicts a user’s preference for an item from historical user–item interactions rather than from explicit descriptions of users or items. Its central intuition is that interaction patterns themselves reveal preference structure: users who have behaved similarly in the past are likely to have similar preferences, and items that receive similar responses tend to appeal to similar users. In practice, these interactions are typically represented as a sparse user–item matrix, where observed entries may correspond to either explicit ratings or implicit feedback such as clicks or purchases [3].

Collaborative filtering methods are commonly divided into neighborhood-based and model-based approaches. Neighborhood-based methods estimate the score of an unseen item from observed interactions involving similar users or similar items. In contrast, model-based methods learn a compact representation of the interaction structure directly from data. Among these, latent-factor methods have become particularly important, as they represent items in a shared low-dimensional space learned from the interaction matrix itself.

In this project, we use a latent-factor item-based variant of collaborative filtering. Item representations are learned from the interaction matrix using truncated singular value decomposition (SVD), producing a low-dimensional embedding for each item. A user profile is then constructed from the embeddings of previously interacted items, with higher weight assigned to more recent interactions. Candidate items are finally scored according to their similarity to this profile. This makes the method collaborative, since it relies entirely on interaction data, while also remaining simple and efficient at inference time.

A major strength of collaborative filtering is that it can capture complex preference patterns without requiring handcrafted item features. However, its main limitation is the cold-start problem: if a user or item has little or no interaction history, the model has limited information from which to infer preferences. Collaborative filtering therefore tends to perform best in warm-start settings, where sufficient historical interaction data are available [2].

2.3 Weighted Hybrid Filtering

Hybrid filtering is a recommendation strategy that combines the outputs of two or more underlying recommenders in an attempt to obtain a stronger system than any of its components in isolation. The motivation is that different recommendation strategies tend to have complementary strengths and weaknesses: collaborative filtering exploits patterns of co-consumption between users but struggles with items that have received little or no interaction, while content-based filtering relies on item features and can therefore handle unseen items, but is limited to recommending articles similar to what a user has already read. By fusing the two, a hybrid can in principle retain the personalization quality of each component while mitigating their individual failure modes [4].

The weighted hybrid computes a score for each candidate article as a linear combination of the scores produced by the underlying models, as shown in Equation 1, where (ω_{cf}) and (ω_{cbf}) are the component weights and norm is a normalization function applied to each component’s scores.

$$S_{final} = \omega_{cf} \cdot \text{norm}(S_{cf}) + \omega_{cbf} \cdot \text{norm}(S_{cbf}) \quad (1)$$

A practical complication is that the raw scores of different recommenders typically live on different scales: the cosine similarities produced by an SVD-based collaborative model, for instance, have a very different range and distribution from the TF-IDF cosine similarities produced by a content-based model. If the two were combined directly, the component with the larger raw magnitudes would dominate the ranking regardless of the weights chosen. To prevent this, the scores of each component are rescaled onto a comparable range by the normalization function in Equation 1 before the weighted sum is taken. The weights themselves control the relative influence of each component in the final ranking and can be tuned on held-out data to emphasize whichever component performs better for a given task and dataset.

3 Exploratory Data Analysis

Designing an effective recommender system requires understanding both the properties of the items and the patterns of user behaviour. The goal of the exploratory data analysis (EDA) is therefore not only to describe the dataset, but to identify characteristics that directly influence the design and expected performance of different recommendation approaches.

3.1 Category Distribution

We first examine the distribution of article categories. As shown in Figure 1, a small number of categories, particularly *news* and *sports*, dominate the dataset, while the remaining categories appear with relatively similar but lower frequencies.

This imbalance introduces a potential bias in recommendation performance. In particular, a model that primarily recommends items from dominant categories may achieve high accuracy by exploiting popularity rather than capturing user-specific preferences. This highlights a limitation of accuracy-based evaluation and motivates the inclusion of beyond-accuracy metrics such as diversity and novelty.

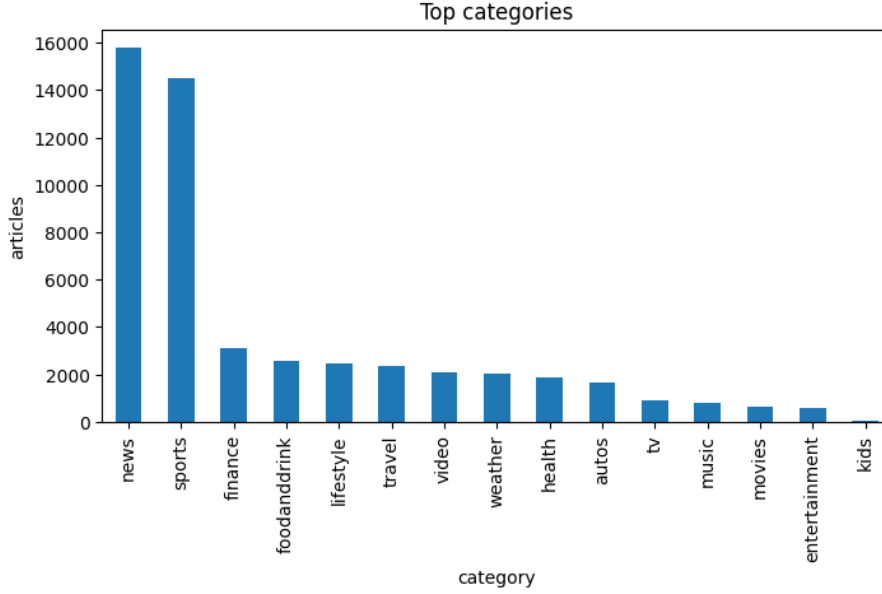


Figure 1: Distribution of article categories in the dataset. A small number of dominant categories account for a large proportion of the articles.

3.2 Article Popularity

We further analyze article popularity by examining the distribution of clicks per item. As illustrated in Figure 2, the dataset exhibits a clear long-tail distribution, where a small number of articles receive a disproportionately large number of clicks.

This reinforces the presence of popularity bias. Without proper control, recommendation models may over-emphasize popular items, reducing exposure to less frequently consumed content. This also explains why a popularity-based baseline can achieve competitive performance despite lacking personalization. To account for this effect, we include a novelty metric in the evaluation to measure whether models recommend less popular items.

3.3 Textual Features

Next, we analyze the textual properties of the articles. Figure 3 shows the distributions of title and abstract lengths.

We observe that titles are consistently short, while abstracts exhibit greater variability, with a non-negligible proportion of articles lacking an abstract altogether. This has direct implications for content-based recommendation. Relying on a single textual field would result in weak or incomplete representations for some items. To mitigate this, we construct item representations by combining multiple sources of information, including title, abstract, category, and subcategory, resulting in more robust feature vectors.

3.4 User Behaviour

Finally, we examine user behaviour through session statistics. As shown in Figure 4, the distributions of impressions, clicks, and history length per session are strongly heavy-tailed.

This indicates that most users interact with only a small number of articles, while a minority of users are highly active. As a result, the user-item interaction matrix is highly sparse, which limits the effectiveness of collaborative filtering methods that rely on sufficient co-consumption signals.

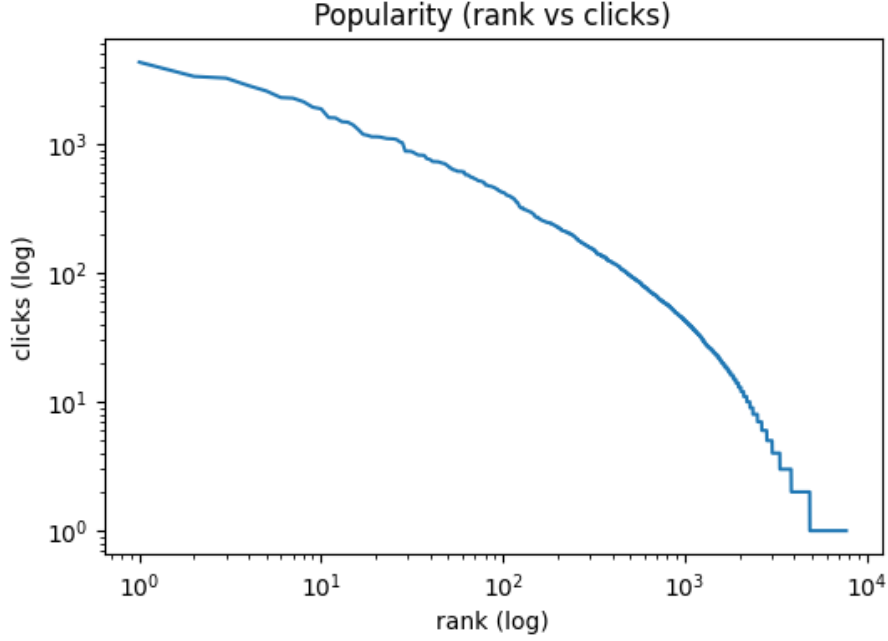


Figure 2: Distribution of article popularity (number of clicks per item), showing a long-tail pattern.

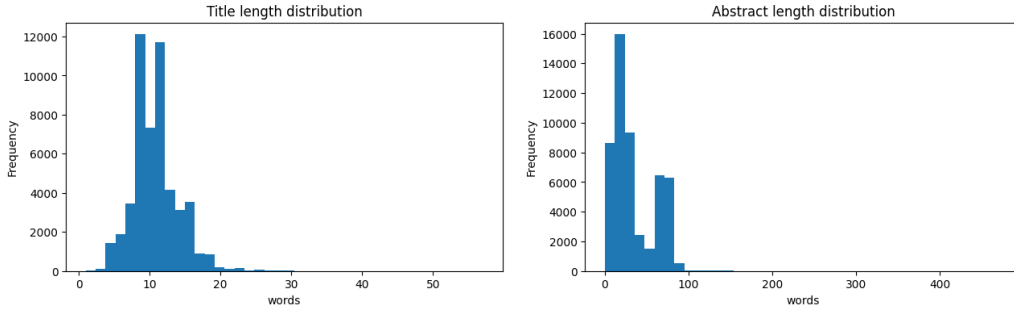


Figure 3: Distribution of title lengths (left) and abstract lengths (right). Titles are consistently short, while abstracts vary more widely in length and are sometimes missing.

This further motivates the use of content-based methods, which are more robust in sparse and cold-start settings.

We also observe temporal patterns in user activity, as shown in Figure 5, where activity peaks during late morning hours and gradually declines throughout the day.

While temporal dynamics are not explicitly modeled in this work, such patterns could be incorporated in future extensions through context-aware recommendation methods.

3.5 Summary of Insights

Overall, the exploratory analysis provides several key insights that directly inform the design of the recommendation models. The sparsity of user interactions motivates the use of content-based methods for handling cold-start scenarios, while collaborative filtering is expected to benefit from larger interaction datasets. The strong influence of popularity signals highlights the need for beyond-accuracy metrics such as novelty and diversity. Finally, the complementary strengths of content-based and collaborative approaches motivate the use of a hybrid model to combine their advantages.

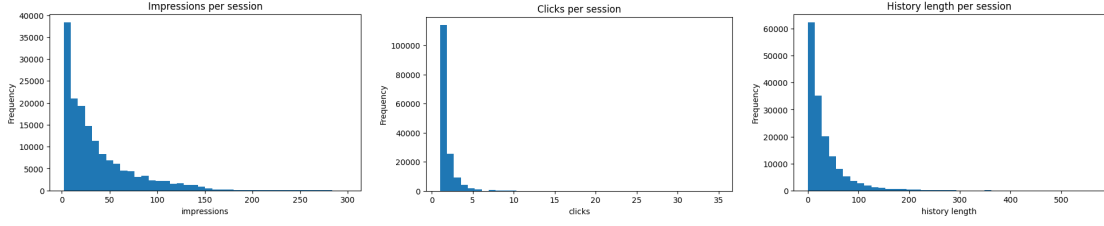


Figure 4: Distribution of user activity statistics (impressions, clicks, and history length per session), showing strongly heavy-tailed behaviour.

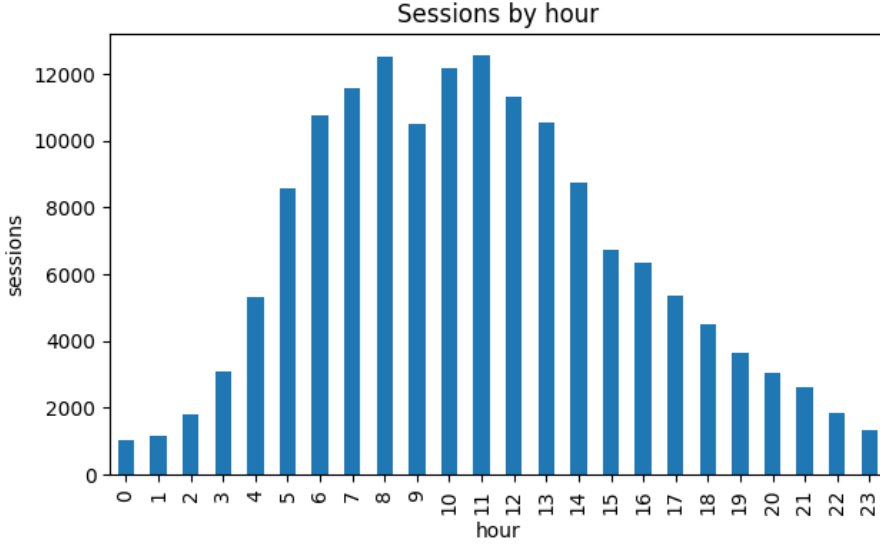


Figure 5: Distribution of session start times, showing daily patterns in user activity.

4 Method

This section describes the implementation of the recommender systems used in the study, along with the evaluation metrics.

4.1 Dataset

The recommender systems were designed and evaluated on the MIND (Microsoft News Dataset) benchmark, a large-scale English-language dataset of user-news interactions collected from anonymized Microsoft News logs. MIND is distributed in two sizes: a *small* version containing roughly 50,000 users and a *large* version with around one million users.

All models were trained on the MIND small training split due to computational constraints. To study the effect of dataset scale on model performance, we additionally report evaluation results on both the MIND small and MIND large development sets. This allows us to analyze how different recommendation strategies behave under varying levels of interaction data.

The dataset is split into a training and a development set, each consisting of two files: **news.tsv**, which describes each article by its ID, category, subcategory, title, abstract, URL, and extracted named entities, and **behaviors.tsv**, which records user impression logs. Each impression contains a user ID, timestamp, prior click history, and a set of candidate articles labeled according to whether they were clicked.

This structure makes MIND well suited for the task, as it provides both the textual content required by the content-based model and the interaction data required by the collaborative model, while the

impression-level click labels provide a natural ground truth for ranking evaluation. The training split was used to fit all models, and the development splits were used as held-out evaluation sets [1].

4.2 Baseline

As a baseline, a simple *most popular* recommender was implemented. Article popularity was computed as the total number of clicks in the training set, and each candidate was scored at inference time using this count.

The model is non-personalized, meaning that recommendations are identical for all users within an impression. Items not observed during training were assigned a score of zero. This baseline serves both as a lower bound on accuracy and as a reference point for beyond-accuracy metrics, as popularity-based recommenders tend to favour frequently clicked but less novel items.

4.3 Content-Based Recommender

The content-based model ranks candidate articles based on their textual similarity to a user profile constructed from reading history. Each article is represented as a TF-IDF vector computed over its title, abstract, category, and subcategory.

A user profile is constructed as a recency-weighted sum of the vectors corresponding to previously read articles, giving higher importance to more recent interactions. Candidate items are then scored using cosine similarity between their vector representation and the user profile. Since this representation depends only on article content, the model is able to score items that were not observed during training, making it suitable for cold-start scenarios.

4.4 Collaborative Recommender

The collaborative model learns from patterns of co-consumption across users. A sparse user-item interaction matrix is constructed by aggregating each user’s reading history and positively clicked articles from the training data.

This matrix is factorized using truncated singular value decomposition (SVD) to obtain low-dimensional latent representations for items. At inference time, a user profile is computed as a recency-weighted average of the embeddings of previously interacted items. Candidate items are then scored using cosine similarity between their embedding and the user profile.

Since item embeddings are learned only for items observed during training, unseen items cannot be scored, which represents a limitation of collaborative filtering in cold-start settings.

4.5 Hybrid Recommender

The hybrid model combines the collaborative and content-based approaches by fusing their scores into a single ranking. The goal is to leverage both interaction-based and content-based signals while mitigating their individual weaknesses.

For each impression, we obtain a score vector from each component model. These scores are normalized onto a comparable range to prevent one model from dominating due to scale differences. The final score is then computed as a weighted linear combination of the normalized scores.

Initial experiments used default weights of 0.35 for collaborative filtering and 0.65 for content-based filtering. However, the final reported results use tuned weights selected on the validation set, where the best performance was achieved with a higher emphasis on the content-based component.

4.6 Metrics

Model performance was evaluated using both accuracy and beyond-accuracy metrics.

For accuracy, we report AUC, MRR, nDCG@5, and nDCG@10. These metrics are computed per impression and averaged across sessions. AUC measures the overall ranking quality, while MRR and nDCG emphasize the correctness of the top-ranked items, reflecting the importance of early positions in news recommendation.

To capture aspects not reflected by accuracy alone, we additionally report beyond-accuracy metrics computed on the top-10 recommendations. Novelty is defined as the mean self-information $-\log_2 p(\text{item})$ under the training popularity distribution, measuring the tendency to recommend less popular items. Diversity is computed as the fraction of distinct categories and subcategories among the top-10 recommendations, measuring topical variety.

Together, these metrics provide a more comprehensive evaluation by capturing both ranking performance and the trade-offs between personalization, novelty, and diversity.

5 Results

This section presents the performance of the four recommender strategies on both the MIND small and MIND large development sets. Each model was evaluated on the same set of impression sessions using the metrics described previously, ensuring that differences in performance are attributable to the recommendation strategy rather than the evaluation setup.

The hybrid recommender combines collaborative filtering (CF) and content-based filtering (CBF). The combination weights were tuned in increments of 0.1, and the best performance was obtained with weights of 0.1 for CF and 0.9 for CBF, indicating that content-based signals dominate in this setting. The tuning was performed on the validation set by selecting the weight combination that maximized AUC.

5.1 Results on MIND Small

Model	AUC	MRR	nDCG@5	nDCG@10	Div/cat	Div/sub	Novelty
Baseline	0.5318	0.2671	0.2460	0.3098	0.6898	0.8790	14.0431
Collaborative	0.5429	0.2632	0.2473	0.3088	0.6774	0.8840	14.5847
Content	0.6073	0.3306	0.3108	0.3718	0.5810	0.7981	16.0303
Hybrid	0.6126	0.3310	0.3112	0.3731	0.5844	0.8017	15.7959

Table 1: Evaluation results for the four recommender strategies on the MIND small development set.

On the MIND small dataset, the content-based model achieves a substantial improvement over both the baseline and collaborative filtering across all accuracy metrics. This is consistent with the observed sparsity of user interactions, which limits the effectiveness of collaborative filtering. In contrast, the availability of textual features enables the content-based model to construct informative item representations.

The hybrid model achieves the strongest overall accuracy, slightly improving over the content-based model. This suggests that collaborative signals provide complementary information, although their contribution remains limited due to sparse interaction data.

Collaborative filtering performs only marginally better than the baseline. This indicates that interaction data alone is insufficient to capture reliable similarity patterns in the small dataset.

5.2 Results on MIND Large

Model	AUC	MRR	nDCG@5	nDCG@10	Div/cat	Div/sub	Novelty
Baseline	0.5385	0.2618	0.2453	0.3079	0.6937	0.8802	14.9585
CF	0.5541	0.2795	0.2577	0.3223	0.6681	0.8733	15.6391
CBF	0.6059	0.3300	0.3097	0.3711	0.5817	0.7977	17.8682
Hybrid	0.6084	0.3272	0.3066	0.3692	0.6079	0.8202	16.5903

Table 2: Performance comparison of recommendation models on the MIND large dataset across both accuracy and beyond-accuracy metrics.

On the MIND large dataset, collaborative filtering shows a more noticeable improvement compared to the small dataset. This highlights the importance of dataset scale for collaborative methods, which rely on sufficient interaction data to learn meaningful patterns.

Content-based filtering remains strong and achieves the highest MRR and nDCG values, indicating that it is particularly effective at ranking relevant items at the top of the recommendation list.

The hybrid model achieves the highest AUC, demonstrating improved overall ranking performance when combining both approaches. Compared to the small dataset, the gap between collaborative and content-based filtering is reduced, further emphasizing the role of data scale.

5.3 Comparison Across Dataset Scales

Comparing results across the two datasets reveals clear differences in model behaviour. In particular, collaborative filtering benefits significantly from increased interaction data, as reflected in its improved performance on the larger dataset.

In contrast, the performance of the content-based model remains relatively stable across both datasets, indicating that it relies primarily on item features rather than interaction data. As a result, the performance gap between collaborative and content-based filtering decreases when moving to the larger dataset, although the content-based model remains the strongest individual approach.

These findings suggest that content-based methods are more robust in sparse settings, while collaborative methods require sufficient data scale to reach their full potential.

5.4 Beyond-Accuracy Trade-offs

Beyond accuracy, the models exhibit clear trade-offs. The baseline achieves relatively high diversity but low novelty, as it tends to recommend popular items across categories. In contrast, the content-based model achieves the highest novelty, indicating a tendency to recommend less popular items, but at the cost of reduced diversity.

The hybrid model provides a more balanced trade-off between accuracy, diversity, and novelty. While it does not maximize any single metric, it achieves consistently strong performance across all metrics, making it a robust overall approach.

5.5 Summary

Overall, the results highlight that model performance depends strongly on dataset characteristics. Content-based filtering is particularly effective in sparse settings, while collaborative filtering benefits from larger datasets. Hybrid methods can combine the strengths of both approaches, although their effectiveness depends on the quality of the underlying signals.

More generally, the results demonstrate that no single model optimizes all evaluation criteria simultaneously, and that recommender systems must be evaluated in terms of both accuracy and beyond-accuracy trade-offs.

6 Discussion

This section interprets the results and relates them to both the design choices of the models and the characteristics of the dataset.

6.1 Baseline

The popularity-based baseline achieves the lowest accuracy among all models, with an AUC only slightly above random performance. This is expected, as the model does not incorporate any user-specific information and relies solely on global popularity.

The beyond-accuracy metrics provide further insight into its behaviour. While diversity is relatively high, indicating that popular items span multiple categories, the model exhibits the lowest novelty. This confirms that the baseline primarily recommends frequently consumed items, reinforcing the presence of popularity bias. This behaviour also explains why the baseline can achieve reasonable accuracy despite lacking personalization.

6.2 Collaborative Filtering

The collaborative filtering model shows only a modest improvement over the baseline in terms of AUC and performs similarly or slightly worse on ranking metrics such as MRR and nDCG. This behaviour can be explained by the sparsity of user interactions, which limits the availability of reliable co-consumption signals. As many users have short interaction histories, the user-item matrix remains too sparse for robust latent factor estimation.

On the MIND large dataset, however, collaborative filtering shows a clearer improvement. This confirms that CF benefits from increased data scale, as more interactions allow the model to better capture user similarity patterns. The performance gap between CF and content-based filtering decreases on the larger dataset, highlighting the importance of data scale for collaborative approaches.

Nevertheless, even with more data, CF does not outperform the content-based model. This suggests that interaction signals alone may be insufficient in this domain, particularly when compared to the strong signal provided by textual content.

6.3 Content-based Filtering

The content-based model achieves a substantial improvement over both the baseline and collaborative filtering across all accuracy metrics. This is consistent with the availability of rich textual features in the dataset, which enable the construction of informative item representations even in the absence of extensive interaction data.

This makes the model particularly robust to sparsity and cold-start scenarios. However, the beyond-accuracy metrics reveal an important trade-off. While the model achieves the highest novelty, indicating reduced reliance on popular items, it exhibits lower diversity. This is a natural consequence of content similarity, as recommendations tend to cluster around a user’s previously consumed topics.

This highlights a fundamental trade-off between strong personalization and topical diversity in content-based approaches.

6.4 Hybrid

The hybrid model achieves the strongest overall performance in terms of AUC, combining the strengths of collaborative and content-based approaches. The optimal weighting (0.1 CF, 0.9 CBF) reflects the relative importance of the two signals in this setting, where content-based information dominates due to data sparsity.

The improvement over the content-based model is modest but consistent, suggesting that collaborative signals provide complementary information when available. This effect becomes more noticeable on the larger dataset, where increased interaction data improves the quality of collaborative representations.

In terms of beyond-accuracy metrics, the hybrid model achieves a balance between diversity and novelty. Compared to the content-based model, it slightly increases diversity while maintaining relatively high novelty, indicating that combining signals helps mitigate the limitations of each individual approach.

However, the limited improvement over the content-based model suggests that the collaborative signal remains relatively weak. As a result, the hybrid model is largely dominated by the content-based component, and the contribution of CF is constrained by the sparsity of the interaction data. This indicates that hybrid approaches are most beneficial when strong collaborative signals are available.

6.5 Comparison to Related Work

The observed results are consistent with prior findings in news recommendation literature, where content-based approaches are often strong baselines due to the rich textual nature of news articles and the prevalence of cold-start problems. Collaborative filtering methods typically require large amounts of interaction data to perform well and are less effective in sparse settings.

Hybrid approaches are commonly reported to outperform individual models by combining complementary signals. However, the relatively small improvement observed here suggests that the benefit of hybridization depends strongly on the quality of the underlying components, particularly the strength of the collaborative signal.

6.6 Limitations

This work has several limitations. First, the collaborative filtering model relies on a relatively simple matrix factorization approach, which may not fully capture complex user-item interactions. Second, the content-based model is based on TF-IDF representations, which do not capture semantic relationships as effectively as modern embedding-based methods. Finally, temporal dynamics and contextual information are not explicitly modeled, despite their potential importance in news recommendation scenarios.

6.7 Summary

Overall, the results demonstrate that dataset characteristics play a crucial role in determining model performance. The sparsity of user interactions and the richness of textual features make content-based filtering particularly effective in this setting, while collaborative filtering benefits from increased data scale but remains limited when interaction data is insufficient.

The hybrid model successfully combines the strengths of both approaches, achieving strong overall performance. However, the relatively small gains over the content-based model highlight that hybridization does not always yield large improvements, particularly when one component dominates. More generally, the findings suggest that in news recommendation settings with sparse

interaction data, content-based methods are likely to outperform collaborative approaches, while hybrid models provide the greatest benefit when both signals are sufficiently strong.

7 Conclusion

This study compared four recommendation strategies on the MIND dataset: a popularity-based baseline, a collaborative filtering model based on truncated SVD of the implicit interaction matrix, a content-based recommender using TF-IDF representations of article text, and a weighted hybrid of the two personalized models. All methods were evaluated on both the MIND small and MIND large datasets using a consistent set of accuracy and beyond-accuracy metrics.

The results show that the hybrid model achieves the highest AUC, indicating the strongest overall ranking performance. However, the content-based model performs best on ranking metrics such as MRR and nDCG, suggesting that it is particularly effective at placing relevant items at the top of the recommendation list. Overall, the hybrid provides a slight improvement over the content-based model when considering multiple metrics jointly. This highlights the importance of evaluating recommender systems using multiple metrics, as different models may excel under different evaluation criteria.

The content-based model emerges as the strongest individual approach, demonstrating that textual similarity between previously read and candidate articles provides a highly effective signal in news recommendation. In contrast, the collaborative filtering model is limited by data sparsity on the small dataset, but shows improved performance on the larger dataset, confirming its dependence on sufficient interaction data.

The hybrid model offers a small but consistent improvement over the content-based approach, indicating that collaborative signals provide complementary information when available. However, the relatively modest gains suggest that hybrid methods are most beneficial when strong collaborative signals are present, which is not fully the case in sparse settings.

Beyond accuracy, the results highlight clear trade-offs between diversity and novelty. While the baseline produces the most diverse but least novel recommendations, the content-based model generates highly novel but more topically concentrated results. The hybrid model achieves a balance between these properties, demonstrating the benefit of combining multiple signals.

Overall, the findings suggest that model performance in news recommendation is strongly influenced by dataset characteristics. In sparse environments, content-based methods are likely to outperform collaborative approaches, while hybrid models provide the greatest benefit when both content and interaction signals are sufficiently informative.

Future work includes exploring richer semantic representations for content, incorporating temporal dynamics, and further improving hybrid strategies to better balance accuracy and beyond-accuracy objectives.

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Appendix

A Code Repository

<https://github.com/TrippelEM/tdt4215-mind-recsys>